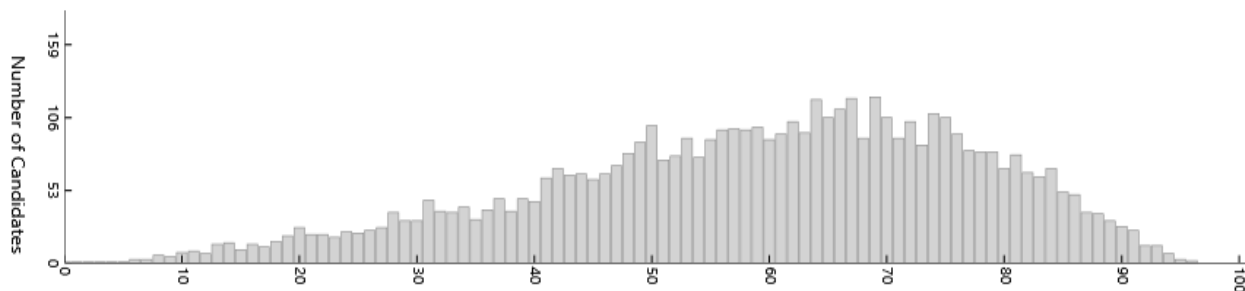




2016 ATAR course examination report: Chemistry

Year	Number who sat	Number of absentees
2016	4997	57

Examination score distribution



Summary

The 2016 examination paper was attempted by 4997 candidates, slightly less than the 5195 candidates in 2015 but maintaining a level above the pre 'half-cohort bubble' of 2014. Chemistry still appears to be one of the most popular of the science, technology, engineering and mathematics (STEM) courses.

The paper as a whole was well received with most candidates managing to attempt the majority of questions. Feedback from examination markers and teachers indicate that the questions reflected a wide range and good spread of the syllabus and that the candidates seemed to have sufficient time to finish the paper. The examination stayed within the boundaries of the syllabus, was considered fair, without surprises and was clear so candidates understood what was being asked. One criticism of the paper was that it did not offer enough challenging questions, requiring higher order thinking or linked concepts and principles from different areas of the syllabus, which would discriminate the very top candidates.

The statistical reports generally support these perceptions. However, the examination mean was 58.73% and the standard deviation was 18.47%. The paper functioned well to discriminate between the ability of candidates with scores ranging from 0.00% to 96.16%.

Section means were:

Section One: Multiple-choice	Mean 16.68(/25)	Max 25.00	Min 0.00
Section Two: Short answer	Mean 19.76(/35)	Max 33.10	Min 0.00
Section Three: Extended answer	Mean 22.39(/40)	Max 39.53	Min 0.00

The mean of 58.73% is an increase on the 2015 paper which had a mean of 55.99%. The mean of section one was 66.72%, up 1.88% from 2015; the mean of section two was 56.46%, up 2.91% and the mean for section three was 55.98%, up 3.35%.

Being the first examination for the new ATAR course, it served to provide a very accessible assessment of the new syllabus material. Many questions were deliberately written in the language of the new syllabus.

General comments

Generally, candidates demonstrated familiarity with most of the concepts and principles of the course but were unable to provide comprehensive explanations, drawing on and linking fundamental chemical principles. Candidates need to realise that marks are not awarded for re-stating information provided in the questions nor for vague references. It was evident within the examination that candidates could often do the calculations well but many did not express themselves clearly and concisely using the appropriate scientific terminology when providing justifications or explaining concepts.

Some markers felt that it was a reasonably difficult paper, especially for the weaker candidates. However, others commented that it was on the easier side with the calculations being very straightforward except for question 42 parts (f) and (g). The perception was that there were only a few questions which could discriminate the top candidates, notably questions 32(b), 34(b), 37 and 39(d).

It was suggested that about 20% of the paper were effectively recall questions rather than applying knowledge. For example, question 26 and question 40(b) combined to give 16 marks on redox chemistry and question 28(b) on an industrial process.

A perception expressed was that the examination paper was very heavily weighted towards organic chemistry. This is probably true as nearly half of the syllabus is specifically organic chemistry or uses organic chemistry as the context in which chemical principles are presented.

It is often problematic in allocating appropriate marks to the different concepts and skills being assessed in examination. This is illustrated by comparing question 31(b) with question 33(b). Question 33(b) was allocated three marks for, what on first appearance is the one concept repeated three times (the successive ionisation of the acidic hydrogen on a carboxylic acid group) yet in order to earn the full three marks candidates needed to:

- recognise, relate and draw corresponding structures
- identify the (–OH group) as being unreactive
- determine that the three (–COOH groups) each in turn lost a proton (H^+) and gained a negative charge.

On the other hand, question 31(b) was also allocated three marks and required three separate and significant chemical concepts.

Candidates did not answer question 35 and question 37 on intermolecular forces well. The syllabus states: *'organic compounds display characteristic physical properties, including boiling point and solubility in water and organic solvents; these properties can be explained in terms of intermolecular forces (dispersion forces, dipole-dipole interactions and hydrogen bonds) which are influenced by the nature of the functional groups'*. This has historically been an area of weakness and remains so. Candidates often fail to recognise that molecules might exhibit more than one type of intermolecular force and that given a large enough molecule, the cumulative dispersion forces are often greater than the other forces present.

Many candidates were not allocated full marks for their answers involving calculations because their setting out was untidy and lacked clarity; many candidates missed critical steps and failed to identify clearly to which species they were referring which resulted in ambiguity. If a mistake was made resulting in an incorrect answer and the mistake cannot be identified, candidates scored lower generally than they otherwise would have done. Full marks cannot be awarded to an answer if it is not shown clearly how that answer was derived.

Often there was a lack of clarity of logic in the steps undertaken to solve a problem, justify a response or explain an answer. The lack of appropriate units being used throughout working and especially for a final answer added to this problem.

As indicated in the examination design brief, the new syllabus demands an understanding of error in determining a quantifiable answer and answers are now expected to be expressed to the appropriate number of significant figures. While all answers should be so expressed, there was only one question in this examination paper where marks were specifically allocated for the appropriate number of significant figures; some candidates did not express the final answer to the appropriate significant figure and so were not awarded this mark.

Another area of concern is the tendency for candidates to use minimal scientific jargon and expression leaving the marker unable to determine whether the candidate did not know or did not take the time to express what they did know.

For example:

- (i) in question 34, many candidates failed to demonstrate how they arrived at the reading off the curve and simply stated a figure to one significant figure. Candidates need to demonstrate:
 - care when reading off the graph in an examination, they need to demonstrate they have used a ruler by showing the relevant markings on the graph rather than just 'eyeball it'
 - an appreciation for the degrees of accuracy available when reading an equivalence point; stating '5.0' or '4.8' rather than '5'.
- (ii) many candidates used lazy expressions simply referring to the ambiguous 'NaOH' rather than 'sodium hydroxide solution' or 'NaOH(aq)'. Strictly speaking, NaOH refers to the compound which is a solid at normal temperature and pressure.
- (iii) many candidates failed to indicate the conditions (catalyst/acidic conditions) for reactions when writing the equations depicting particular processes.

An area that distinguished candidates of different abilities was their understanding of the:

- process involved in the buffering action in a buffering system
- inter and intra-molecular forces attraction causing B-pleating.

As has always been the case, the examination markers seek to award all the marks possible to the candidate but this is not possible if the candidates' responses cannot be read because their writing is illegible or the pens or pencils used are too light. If the candidates' work cannot be read, they will potentially miss out on marks.

Advice for candidates

- Be aware that the examinable content of the syllabus includes knowledge, understandings and skills.
- Be conversant with the terms used in the syllabus, e.g. chemical test, data, evidence, genre, hydrolysis, hypothesis, ionisation energy, law, measurement error, origin of intermolecular interactions, partial pressures, polarities, polymerisation, pyramidal, risk assessments, research ethics, significant figures, systematic error, theory, trend, uncertainty etc.
- Understand the differences between similar terms: e.g.
 - intermolecular force and intramolecular force
 - equivalence point and end point.
- Practise writing clear, concise and coherent explanations and justifications; incorporating illustrative, labelled diagrams that are clear and relevant.
- Make it a habit to use the appropriate units and express numerical answers to the appropriate significant figures.

- Be familiar with the content of, and how to use, the Chemistry Data Booklet; knowing where to find the information it contains quickly.
- Know how to generate, read and apply data from graphs and tables.
- Use the given formula to determine the nature and bonding of a substance and all its intermolecular forces.
- Be familiar with, and how to clearly express, the expected observations and inferences that can be made for the reactions indicated in the syllabus. For example: All solutions are clear; some are colourless and some exhibit a colour. Referring to clear solution is redundant and is not the same as referring to a colourless solution. Copper sulfate solution is blue, while sodium sulfate solution is colourless; both are clear because they are solutions.
- Practise writing equations providing the appropriate formula and state symbols for only those species that are taking part in the reaction unless otherwise directed.
- Answer questions with specific reference to what the question demands rather than in general terms or by implication. Candidates are not necessarily awarded any marks for correct chemistry unless it pertains specifically to the question asked.
- Read and select the relevant information provided carefully in order to answer the specific question asked.
- Be prepared for the examination with all the authorised writing materials you might need, choosing pens and pencils which are easy to use and to read; have spares.

Advice for teachers

- Ensure coverage of all syllabus dot points as set out in the syllabus document. The entire syllabus is not necessarily examined in any one examination but could be expected to be assessed in its entirety over the course of a few years. Some syllabus points are examined regularly as they are essential for the understanding and communication of chemical concepts and processes.
- Keep the Learning Outcomes clearly in mind, the knowledge, understandings and skills examinable are found in the Unit Content. This includes:
 - Science Inquiry Skills
 - Science as a Human Endeavour
 - Science Understanding
- Note that Le Châtelier's Principle is a predicting tool not an explanation. Distinguish between the use of:
 - collision theory to explain the changes in rates of reaction
 - the change in rates of the forward and reverse reactions to determine and explain the effect on equilibrium position
 - Le Châtelier's Principle to predict the impact of changing conditions to a system on its chemical equilibrium.
- Engage students in conducting relevant experiments and problem solving. This year, candidates struggled to apply knowledge of common laboratory procedures.
- Insist that students use the appropriate units and express numerical answers to the appropriate significant figures unless otherwise directed.
- Provide students with practice in writing extended answers, justifications and explanations; incorporating illustrative diagrams that are clear, labelled and relevant.
- Encourage students to set out their working and reasoning clearly; full marks are not awarded unless it is clear how the answer was obtained and the relevant chemical ideas (such as relevant mole ratios) are shown.

Comments on specific sections and questions

Section One: Multiple-choice

Attempted by 4997 candidates

Mean 16.68(/25)

Max 25.00 Min 0.00

The easiest questions, with means of 94% and 86% respectively were questions 23 and 19.

- Question 23 required candidates to recognise the process of fermentation from a flow chart provided.
- Question 19 required candidates to recognise the products of simple organic reactions.

Another seven questions had means of 80% and above; these included questions 1, 8, 15 and 16. Most of the more capable candidates did well with these questions.

- There was some criticism of question 1 as the addition of an inert gas in a closed system required that constant volume or constant pressure also be stated. This did not seem to have negatively impacted candidates' capacity to answer this question correctly.

The most challenging questions, with means less than 50%, were questions 3, 6, 20 and 22.

- Question 3 required candidates to select the correct statement concerning a given equilibrium system.
- Question 6 required candidates to identify the set of data that indicated a systematic error; this aspect of experimental error is new to the Chemistry syllabus.
- Question 20 required candidates to identify the compounds that produce a polymer. The alternative given as the correct answer included four compounds, including compound III ($\text{CH}_2=\text{CHOH}$). This compound has the ability to slowly tautomerise to the aldehyde ($\text{CH}_3\text{CH}=\text{O}$) and in that state could not be used to produce a polymer. At room temperature, acetaldehyde ($\text{CH}_3\text{CH}=\text{O}$) is more stable than the vinyl alcohol ($\text{CH}_2=\text{CHOH}$). The tautomerisation occurs slowly normally but is catalyzed by acids. Photo-induced keto-enol tautomerisation is also viable under atmospheric conditions. Candidates were unlikely to know this and hence answer (c) was chosen as the correct answer. Many candidates chose answer (a) which does not include structure III.
- Question 22 required candidates to recognise the relative hardness of water samples according to the volume of soap required to produce a permanent lather. The distractors of boiling point and mass of a precipitate produced caused confusion for candidates; making them very effective distractors.

Question 01 attempted by 4997 candidates	Mean 0.80(/1)	Max 1	Min 0
Question 02 attempted by 4997 candidates	Mean 0.79(/1)	Max 1	Min 0
Question 03 attempted by 4997 candidates	Mean 0.41(/1)	Max 1	Min 0
Question 04 attempted by 4997 candidates	Mean 0.62(/1)	Max 1	Min 0
Question 05 attempted by 4997 candidates	Mean 0.68(/1)	Max 1	Min 0
Question 06 attempted by 4997 candidates	Mean 0.44(/1)	Max 1	Min 0
Question 07 attempted by 4997 candidates	Mean 0.52(/1)	Max 1	Min 0
Question 08 attempted by 4997 candidates	Mean 0.80(/1)	Max 1	Min 0
Question 09 attempted by 4997 candidates	Mean 0.73(/1)	Max 1	Min 0
Question 10 attempted by 4997 candidates	Mean 0.73(/1)	Max 1	Min 0

Question 11 attempted by 4997 candidates	Mean 0.73(/1)	Max 1	Min 0
Question 12 attempted by 4997 candidates	Mean 0.75(/1)	Max 1	Min 0
Question 13 attempted by 4997 candidates	Mean 0.64(/1)	Max 1	Min 0
Question 14 attempted by 4997 candidates	Mean 0.47(/1)	Max 1	Min 0
Question 15 attempted by 4997 candidates	Mean 0.81(/1)	Max 1	Min 0
Question 16 attempted by 4997 candidates	Mean 0.80(/1)	Max 1	Min 0
Question 17 attempted by 4997 candidates	Mean 0.54(/1)	Max 1	Min 0
Question 18 attempted by 4997 candidates	Mean 0.70(/1)	Max 1	Min 0
Question 19 attempted by 4997 candidates	Mean 0.86(/1)	Max 1	Min 0
Question 20 attempted by 4997 candidates	Mean 0.41(/1)	Max 1	Min 0
Question 21 attempted by 4997 candidates	Mean 0.63(/1)	Max 1	Min 0
Question 22 attempted by 4997 candidates	Mean 0.44(/1)	Max 1	Min 0
Question 23 attempted by 4997 candidates	Mean 0.94(/1)	Max 1	Min 0
Question 24 attempted by 4997 candidates	Mean 0.77(/1)	Max 1	Min 0
Question 25 attempted by 4997 candidates	Mean 0.66(/1)	Max 1	Min 0

Section Two: Short answer

Attempted by 4990 candidates Mean 19.76(/35) Max 33.10 Min 0.00

Question 26 attempted by 4925 candidates Mean 3.64(/6) Max 6 Min 0
 Part (a) - Many candidates indicated for both cells that whilst an electrolyte is required, many stated that it was (an aqueous) solution; few indicated that the electrolyte in electrolytic cells can be molten. Part (b) - Candidates did not indicate clearly to which cell they were referring when describing differences and hence missed out on marks.

Question 27 attempted by 4771 candidates Mean 4.17(/9) Max 9 Min 0
 Part (a) was done very poorly with many candidates writing vague and incomplete observations; many wrote inferences (the substances/functional groups) rather than observations. Many candidates seemed to know the reactions but stated inferences not observations and hence missed marks. This question required a response to the appearance of the reactants and the products as well as observations of changes but when writing observations, many candidates merely described the product(s). Candidates need to be reminded of the difference between clear and colourless solutions. Many candidates referred to solutions as being 'clear'. It is unacceptable to use clear in place of 'colourless'. Like all solutions, acidified potassium permanganate solution is clear but not colourless. Candidates also failed to recognise the significance of the term excess in parts (ii) and (iii) where respectively, the white solid would completely disappear (dissolve) and the purple colour would not just fade/discolour but become the faint pink. Part (b) - Many candidates did not attempt the equation and simply drew the structure of pentanoic acid.

Question 28 attempted by 4569 candidates Mean 2.06(/6) Max 6 Min 0

Part (a) - The question asks candidates to 'Draw a condensed structure ... containing the same number of carbon atoms in the chain'. Candidates interpreted this differently; some as a total of 16 carbons while others as the intended 16 carbons plus the methyl group. The marking key was changed to accept either. Nearly one-fifth of the candidates did not attempt this question. One could speculate that being a new concept presented for the first time in this course whether it was something that all candidates were taught. The ester structure was often given the wrong way around and many candidates drew a hydrocarbon, failing to realise that it was an ester and consequently lost a potential three marks. Some candidates drew the structure as an ether rather than ester or were ambiguous in drawing ie:

–COOC–R or –COC–R–O. Part (b) - It was difficult to determine if the candidate response was adequate for the marks as candidates simply stated some benefits of the lipase method and did not provide an argument (as asked in the question) to justify its use over base-catalysed method. Many answered this as listing advantages of catalysed reactions, stating or assuming the 'NaOH method' is not a catalysed method, even though it is stated 'base-catalysed method' in the question. Correct interpretation of this question often resulted in responses that were too simple. For example, candidates stated that:

- it 'doesn't produce soap' but failed to make the link to an increased yield or contamination needing to be removed
- lipase is a better choice but did not justify their argument.

Many candidates failed to 'justify' their answer.

Question 29 attempted by 4916 candidates Mean 5.97(/9) Max 9 Min 0

Parts (a), (b) and (c) allowed for a variety of possible answers. Each response that did not correspond to an example given in the marking key often had to be validated and 'researched' by each marker in order to award a mark. Part (c) - Although candidates stated some properties of their polymer, the property did not always relate to the use given.

Part (d) - Candidates need to ensure that the bonding connections are correct when drawing the structure. e.g. HO-C not OH-C. Part (e) - This question required the specific term 'β-pleated sheets' as stated in the syllabus. Many candidates submitted similar answers indicating they understood the chemistry of the question, such as pleated sheets.

Question 30 attempted by 4588 candidates Mean 2.91(/5) Max 5 Min 0

Part (a) - Although taken directly from the syllabus statement, some candidates interpreted the question as requiring the oxidation of ethanol to ethanoic acid as there was no explicit mention of ethanoic acid being available for the second step. There was no penalty for doing this step but it did mean that the candidates had difficulty with part (b). Some 23% of candidates did not attempt part (b). The most common error was omitting the water in the esterification reaction.

Question 31 attempted by 4944 candidates Mean 3.35(/6) Max 6 Min 0

Part (b) - Most candidates could write the equation but few referred to the Brønsted–Lowry Model and even fewer compared the concentration of hydroxide ions and hydrogen ions. Most simply made a statement such as, 'since hydroxide ions are produced, then the solution is basic'. Candidates need to discuss pH in terms of comparing the concentrations of hydroxide and hydrogen ions.

Question 32 attempted by 4780 candidates Mean 2.67(/6) Max 6 Min 0

Part (a) - Most candidates correctly wrote the equilibrium equation with the appropriate double arrow. Part (b) - This question discriminated the top candidates from the rest. Many candidates gave a general description of what a buffer does without relating any of what they wrote to the actual question; there are no marks for this. Candidates failed to link addition of HCl(aq) to an increase in $[\text{H}_3\text{O}^+]$; many simply made a statement such as, 'there are more hydrogen ions'. Very few could relate the difference in water with the buffer system to the difference in concentration of the conjugate bases. Many others simply said that, 'water was not a buffer'. They needed to go on to say why it is not a buffer. Many candidates tried to use Le Chatelier's Principle as an explanation; this is a predictor and cannot be used as an explanation medium.

Question 33 attempted by 4916 candidates Mean 3.13(/5) Max 5 Min 0

Part (a) - The question should have said 'to completion' or 'in excess acid' in order to illicit the desired outcome as many candidates did not take the reaction to completion. A common mistake was the missing of water as a product. Part (b) - Few candidates earned part-marks for this question; they could either do this or not. A common error was simply missing the $-\text{OH}$ group in the drawing.

Question 34 attempted by 4974 candidates Mean 3.39(/6) Max 6 Min 0

Part (a) - Many candidates did not follow the instructions provided to label the equivalence point on the curve. In many cases, there was a dot or just some mark on the curve but no label. Few candidates illustrated that they used a ruler to read the corresponding y axis in order to obtain their quoted figure. Part (b) - Candidates only discussed bromocresol green and not the other indicators. Candidates had difficulty in providing a substantial justification worthy of three marks; they often just repeated the fact that the end-point coincided with the equivalence point without possibly knowing why this is so important.

Question 35 attempted by 4939 candidates Mean 5.84(/9) Max 9 Min 0

The question asks to see **all** atoms and **all** bonds for an organic structure drawn. Many candidates drew condensed structures for organic compounds and so missed marks by not drawing all bonds. Many candidates were able to draw the structural formula correctly but were unable to identify all the intermolecular forces present. Some candidates gave an amine for the amide. A number of candidates gave Fl as the symbol for fluorine.

Question 36 attempted by 4963 candidates Mean 8.90(/11) Max 11 Min 0

Part (a)(i) - Many candidates circled the CONH group and whilst the mark was awarded for this, the question specifically asked for the circling of peptide bond. Part (a)(ii) - Most candidates who attempted this question were able to do so accurately. Part (c) - Some candidates circled the groups without naming them and hence missed three marks. Many candidates did not circle the H in the aldehyde group, not recognising that the H is part of the functional group in order to differentiate it from a ketone. Part (d) - The most common error was drawing the structure of the provided isomer of alanine rather than another isomer as required.

Question 37 attempted by 4891 candidates Mean 1.89(/5) Max 5 Min 0

This was a question which provided the top candidates an opportunity to demonstrate their ability. Many candidates explained the difference in boiling points simply in terms of H-bonding (pentanoic acid) being stronger than dipole-dipole forces (pentanal) being stronger than dispersion forces (pentane) without referring to the sum of the intermolecular forces. The candidates needed to emphasise that dispersion forces can effectively be the strongest intermolecular forces for a molecule and it is the additional dipole-dipole forces and H-bonding that explains the differences. Few candidates identified that dispersion forces existed in all compounds but those that did, failed to mention that the strength of these dispersion forces would be similar in each compound.

Section Three: Extended answer

Attempted by 4982 candidates

Mean 22.39(/40)

Max 39.53 Min 0.00

Question 38 attempted by 4939 candidates Mean 9.76(/14) Max 14 Min 0

Part (a) - Only half the candidates could indicate that enzymes are protein molecules which act as biological catalysts, as defined in the syllabus. Part (b) - The use of the term 'charged molecule' caused confusion for some candidates as both the original molecule and the choline are charged. In part (d) candidates often referred to 'the charged molecule' but did not make it clear to which molecule they were referring. Some candidates spoke in general terms rather than specifically to provided reaction. A common error in the calculation was relating the mole ratio of hydrogen to water as 1:2 rather than 2:1. Some candidates left the -OH group off choline or represented the connection incorrectly i.e. OH-C rather than HO-C. Part (c) - This was a standard empirical formula question without any significant or common issues. Part (d) - Few candidates related the empirical formula to choline itself or they failed to state the formula of choline to show that both formulae were identical. More than 28% of candidates did not attempt this part of the question.

Question 39 attempted by 4955 candidates Mean 9.11(/16) Max 16 Min 0

Part (a) - Some candidates failed to specify to which species the n, c or v values they were using referred. Some candidates expressed their working to several significant figures and then recorded their final answer to three significant figures rather than the appropriate number of significant figures as required by the examination design brief. Part (b) - Many candidates simply stated the general role of a catalyst without giving specific justifications for this industrial process. The question asked for two justifications and indicated that stating that a catalyst increases reaction rate or lowers energy of activation would not gain marks. Part (c) - More than 90% of candidates answered this question correctly. Part (d) - This was considered a question which could discriminate the top candidates. Many candidates who responded referred to Le Chatelier's Principle despite the question explicitly stating that the Collision Theory be used to explain the effect of raising the total pressure on the yield. A common mistake many candidates made was to refer to 'more collisions' rather than referring to 'increased rate/frequency of collisions'. Although this type of question has been asked frequently, candidates are still not comparing the forward and reverse reaction rates within the Collision theory to explain the change of equilibrium position.

Question 40 attempted by 4870 candidates Mean 4.85(/11) Max 11 Min 0

Part (a) - Many candidates wrote the equation for the reduction of Cu^{2+} rather than Cu^+ . Some candidates assumed, rather than determined the equation; this might indicate a lack of appreciation for different oxidation states possible. Part (b) - This question is purely recall of a specific syllabus point. Some candidates wrote very good and comprehensive responses. A common mistake was the drawing of an electrochemical cell rather than electrolytic cell or the confusion of elements of both cells, such as drawing two half cells with a salt bridge and then trying to explain an electrolytic process in this cell. Some candidates incorporated a half cell involving Zn or used SO_2 in the half equations $\text{S}^{2-} + \text{O}_2 \rightarrow \text{SO}_2 + 2\text{e}^-$. Candidates either omitted writing about the impurities or were not specific enough in terms of discussing the 'more reactive metals' and 'less reactive metals'.

Question 41 attempted by 4926 candidates Mean 10.74(/18) Max 18 Min 0

Many candidates failed to label either graph resulting in a loss of marks. Graphing skills in general were poor; curved lines were not smooth, straight lines were not straight and horizontal. Many candidates did not demonstrate a strategy to plan and draw the curves and some displayed a poor understanding of what was happening within the system. Part (b) - Many candidates indicated they erroneously thought that K changes with pressure. Parts (c)(i) and (c)(ii) were difficult to mark as many candidates did not indicate clearly the higher temperature graph distinct from the lower temperature one. Part (c)(i) - The main errors were the two curves not meeting at point A and the horizontal lines drawn apart from each other rather than occupying the same space to indicate identical rates. Many candidates failed to commence the reverse reaction curve at zero. Part (c)(ii) - Candidates failed to show that equilibrium would be reached earlier and although the graphs often indicated that rates would be higher they were not drawn correctly. Part (d) - A common omission was not showing the 1:2 ratio of $N_2:O_2$ and both curves starting at the same point. Part (e) - A common omission was the decrease in concentration for all gases. Part (f) - Many candidates wrote that the value for K had to be less than one but this is insufficient; less than one could be 0.9999. The value has to be very small. Some candidates wrote very good responses showing good reasoning.

Question 42 attempted by 4933 candidates Mean 13.61(/26) Max 26 Min 0

Part (a) - Candidates needed to show all their workings to be awarded marks in calculations; this was not always done. Even when calculating the correct answer, it is necessary for candidates to show how they arrived at the answer. Part (b) - Too many candidates simply stated that NaOH was unstable in air - candidates must be more specific in their responses. For example; that sodium hydroxide ($NaOH(s)$) reacts with the carbon dioxide gas ($CO_2(g)$) and/or absorbs moisture from the air. Part (c) - Using the expression ' $c_1V_1=c_2V_2$ ' is not sufficient working to gain full marks unless candidates have shown clearly what each term represents and stated the relationship between NaOH and HCl. Part (d) - Common errors referred to:

- 'rainwater' rather than stating 'the diluted rainwater'
- 'water' rather than stating, 'distilled water'
- 'NaOH' rather than stating 'NaOH(aq)' or 'sodium hydroxide solution'.

Part (e) - A common error was not giving the answer to two decimal places. Part (f) - Poor setting out of the calculation often made it very difficult to follow and award marks. Part (g) - More than one quarter of candidates did not attempt this part of the question. Many candidates who answered the questions ignored the pH 5.6 information and performed the calculation by using the pH of 4.